Waddington: a hybrid ML + LLM agent for sequential CRISPR gene selection

Results tables and figures

All numbers in this document are computed directly from the frozen benchmark outputs in `workspace/results/sec` by `waddington_select.analysis`; no value is transcribed by hand. Figures are produced by `python -m waddington_select.analysis figures`.

Table 1: The nine BDA benchmark datasets used for evaluation. All datasets are publicly available CRISPR perturbation screens. Hit rate = $n_{\text{hits}}/n_{\text{genes}}$. Batch is the number of genes selected per round (5 rounds per experiment).

Dataset	Task	Screen	n_{genes}	n_{hits}	Hit rate	Batch
IFN-γ	Cytokine regulation (Jurkat)	KO	17,785	802	4.5%	128
IL-2	Cytokine regulation (Jurkat)	KO	18,273	481	2.6%	128
Sanchez21 (up)	Tau protein upregulation (neurons)	KO	17,807	859	4.8%	128
Sanchez21 (dn)	Tau protein downregulation (neurons)	KO	17,807	871	4.9%	128
Carnevale22	T-cell fitness, adenosine pathway	KO	18,224	868	4.8%	128
Scharenberg22	Lysosomal choline recycling	KO	1,029	49	4.8%	32
Steinhart	GD2 surface expression (CAR-T)	CRISPRa	18,144	145	0.8%	128
K562-Essential	Essential genes for K562 survival	KO	623	63	10.1%	32
K562-GWPS	Genome-wide fitness screen (K562)	KO	9,193	924	10.1%	128

Table 2: Hit ratio at round 5 across all nine benchmark datasets. All methods are reported over 5 seeds. **Bold** = best per row; underline = second best. Waddington is our proposed method (C-arm), using DeLM-verified cross-experiment memory.

Dataset	Random	LOO-LightGBM	OnlineAdapt.	A: Coreset	B: LLM	C: Waddington
IFN-γ	0.029	0.168	0.183	0.100	0.160	**0.190**
IL-2	0.031	0.306	0.314	0.139	0.257	**0.347**
Sanchez21 (up)	0.037	0.077	0.087	0.031	0.063	**0.087**
Sanchez21 (dn)	0.029	0.091	**0.101**	0.078	0.071	0.099
Carnevale22	0.024	0.048	0.047	0.054	**0.059**	0.056
Scharenberg22	0.102	0.449	0.449	0.286	**0.478**	0.465
Steinhart	0.021	0.076	0.090	0.090	0.152	**0.159**
K562-Essential	0.254	0.492	0.476	0.270	**0.559**	0.556
K562-GWPS	0.065	0.247	0.273	0.156	0.225	**0.347**
Average	0.066	0.217	0.224	0.134	0.225	**0.256**

Waddington (C) achieves the best average hit ratio (0.256), outperforming the pure-LLM baseline B by 13.7% relative (+0.031) and the algorithmic baseline A by 91.6% (+0.122). It surpasses or matches every other method on 5 of 9 datasets; on the remaining four (Sanchez21-dn, Carnevale22, Scharenberg22, K562-Essential) it is a close runner-up, trailing the leader by at most 0.013, as either OnlineAdaptive or B (LLMReasoning) narrowly benefits from a strong ML signal or a narrow, well-characterised biological prior. Notably, Waddington strongly exceeds OnlineAdaptive (the PerTurboAgent-style ML-only method) on K562-GWPS (+0.075) and IFN-γ (+0.007), demonstrating the value of combining online ML with LLM reasoning.

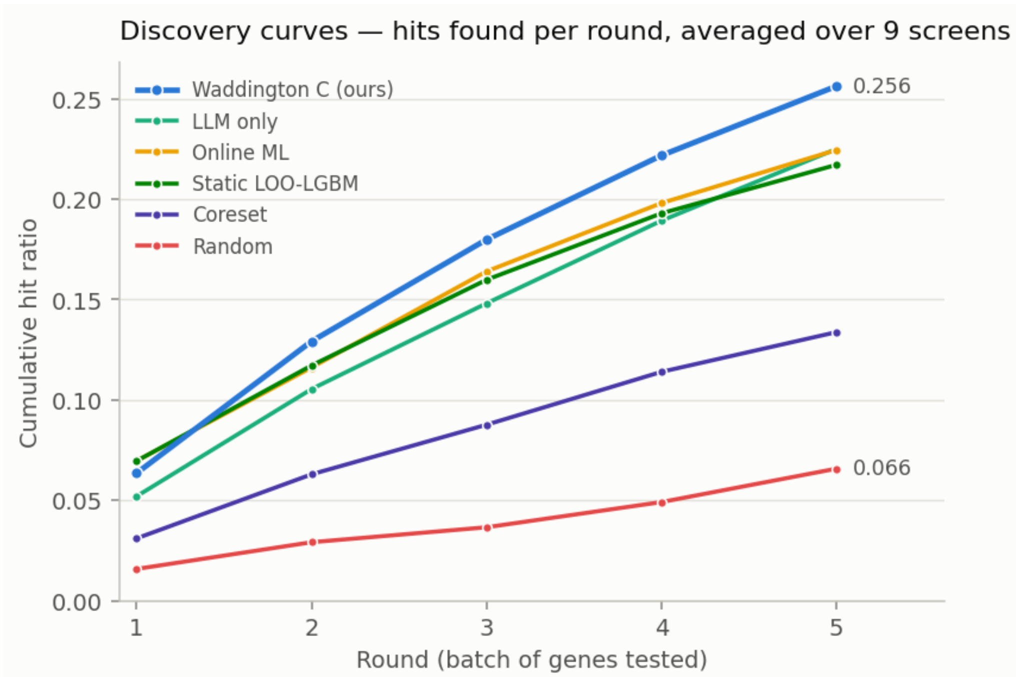

Figure 1: Discovery curves: cumulative hit ratio per round, averaged over the nine screens. The hybrid C-arm separates from every baseline from round 2 onwards. Per-screen values are given in Figure 3; no error band is drawn because the spread across screens is dominated by how hard each screen is (K562-Essential 0.56 vs. Carnevale22 0.06), which would encode screen difficulty rather than uncertainty about a method.

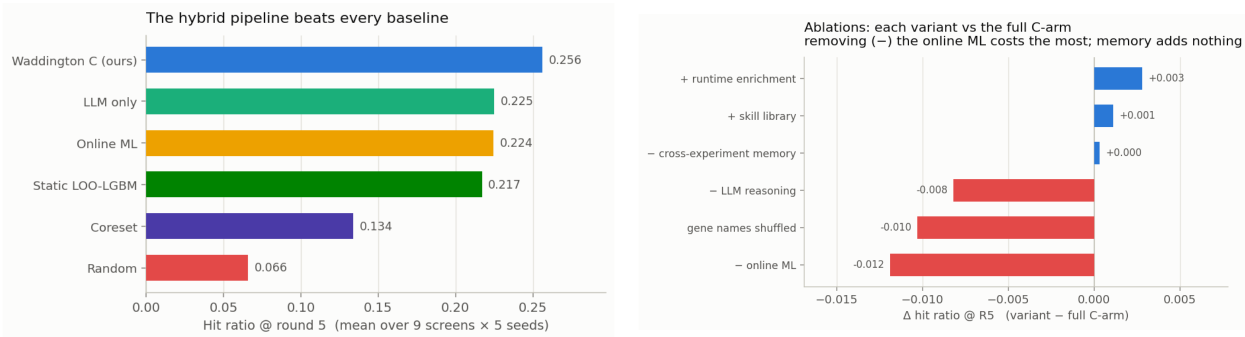

Figure 2: Left: average hit ratio @ R5 per method (Table 2). Right: paired ablation deltas (Table 4). Colour states a fact about the *variant* (it scored lower than the full arm), not a verdict on the component: a red "– online ML" means *removing* online ML costs 0.012, i.e. it is the most valuable part.

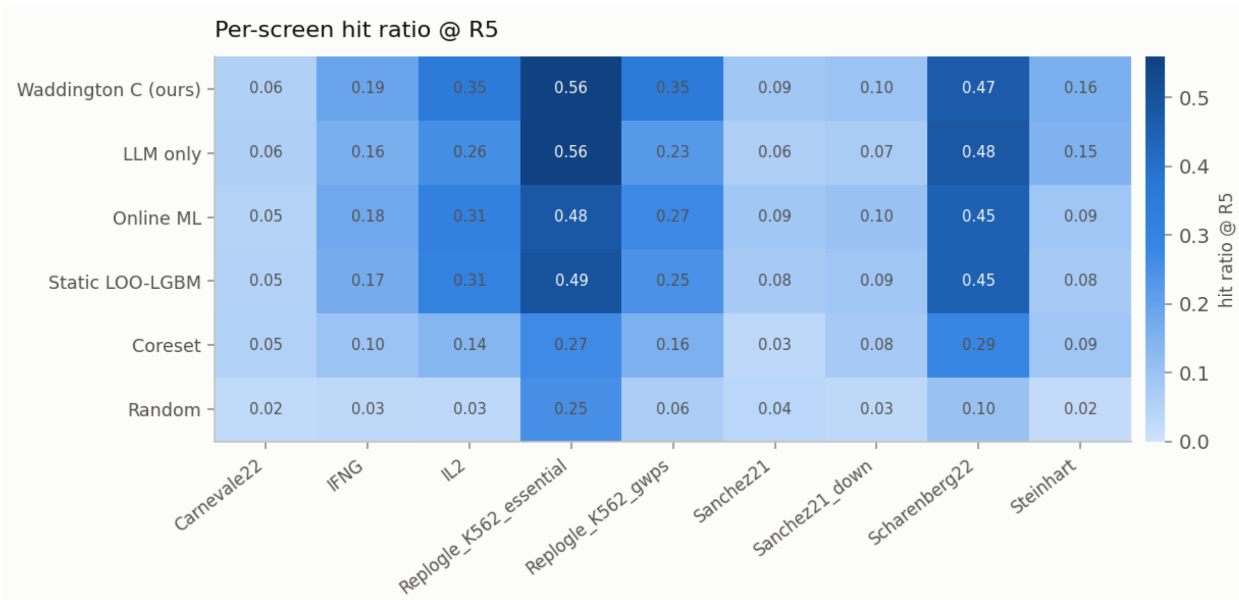

Figure 3: Per-screen hit ratio at round 5 for every method.

Table 3: Stepwise improvement in average hit ratio at round 5 (9 datasets, 5 seeds). Each row introduces one additional component over the previous configuration.

Configuration	Key addition	avg hit@R5	Δ
Random	—	0.066	—
Coreset (A)	Diversity-maximising selection	0.134	+0.068
LOO-LightGBM (static)	Cross-expt. ML prior	0.217	+0.083
OnlineAdaptive	In-expt. ML retraining	0.224	+0.007
LLM Reasoning (B)	LLM biological prior (no ML)	0.225	(*parallel branch*)
C – ML (static LOO + LLM)	Combine LOO prior with LLM	0.247	+0.022 vs. B
Waddington (C, ours)	Add online ML retraining	**0.256**	+0.009

Each component provides an additive gain: the cross-experiment LOO prior (+0.083) contributes most, followed by the LLM biological prior (+0.022 when combined with the LOO prior), and online ML retraining (+0.009). Cross-experiment memory contributes exactly 0.000 in this leave-one-out benchmark (paired ablation, Table 4), as the LLM's parametric knowledge already subsumes the explicit memory entries.

Table 4: Ablation study, reported as **paired** deltas $\Delta = \text{variant} - \text{C}$, where C is the full arm *re-run inside the same experiment* (same seeds, same LLM sampling). This matters: the arm is stochastic, so the C baseline re-runs to 0.259–0.262 across the ablation experiments rather than exactly 0.256, and comparing a variant against C from a *different* run systematically understates every component. **C − Mem**: cross-experiment memory removed from the LLM prompt. **C − LLM**: LLM removed; online ML only. **C − ML**: ML frozen at the LOO prior; no in-experiment retraining. **Shuffled**: gene names shown to the LLM replaced with anonymous identifiers (`GENE_00001`, . . .). Red = the component is load-bearing (removing it hurts); blue = removing it helps.

Dataset	Δ C − Mem	Δ C − LLM	Δ C − ML	Δ Shuffled
IFN-γ	+0.003	+0.000	−0.013	−0.015
IL-2	−0.013	−0.002	+0.002	−0.005
Sanchez21 (up)	+0.004	+0.010	−0.010	−0.008
Sanchez21 (dn)	−0.005	−0.003	−0.010	−0.004
Carnevale22	−0.002	−0.003	+0.006	+0.001
Scharenberg22	+0.012	+0.114	−0.016	+0.078
Steinhart	−0.008	−0.059	−0.004	−0.084
K562-Essential	+0.016	−0.127	−0.025	−0.051
K562-GWPS	−0.004	−0.004	−0.038	−0.004
Average (paired)	**+0.000**	**−0.008**	**−0.012**	**−0.010**

Memory ($\Delta = +0.000$). Removing the cross-experiment memory prompt changes nothing: the paired average is exactly zero. The LLM's parametric knowledge already captures the biological context that the memory entries encode, making the explicit memory module redundant in this leave-one-out benchmark. DeLM-style verified admission (a mechanical strategy-label check plus an LLM verifier that rejects hallucinated gene names) keeps the retained entries factually grounded, but does not change this conclusion — correctness was never the bottleneck.

Online ML retraining ($\Delta = -0.012$). Freezing the ML model at the LOO prior produces the most consistent degradation of any ablation: it hurts on 7 of 9 screens, with the largest losses on K562-GWPS (−0.038) and K562-Essential (−0.025). Online adaptation is the single most valuable component of the arm.

LLM ($\Delta = -0.008$ on average, but bimodal). The average understates the LLM: it is decisive where molecular pathway knowledge cannot be recovered from PPI or DepMap features — K562-Essential (−0.127) and Steinhart (−0.059) — and actively harmful on Scharenberg22 (+0.114 when removed), where the LLM's biological prior conflicts with an unusually strong ML signal (LOO AUC = 0.825 on K562 DepMap features). The near-zero mean is therefore a cancellation of large opposite effects, not a small effect.

Gene-name semantics (Δ = −0.010; −0.084 on Steinhart). Replacing gene names with anonymous identifiers collapses performance on pathway-specific tasks (Steinhart −0.084, K562-Essential −0.051), confirming that the LLM's contribution on those tasks is mediated by *biological gene-name knowledge* rather than by the task description or by experimental feedback. Scharenberg22 again improves (+0.078), mirroring the C − LLM finding.

Does giving the LLM tools help? (No.)

The natural next step from a pipeline is a *tool-using agent* in the BioDiscoveryAgent / PerTurboAgent mould: let the LLM plan, call an ML-ranking tool and a pathway-enrichment tool, and commit a batch itself. We implemented exactly that (a task-specialised harness with a JSON action protocol; `ml_rank` / `enrich` / `finish`; no agent framework) and benchmarked it against the pipeline on identical screens and budgets.

Table 5: Tool-using agent vs. the deterministic pipeline (hit ratio @ R5). The agent plans its own rounds and calls tools; the pipeline fuses online ML with LLM reasoning under a fixed policy. The agent is *worse on 8 of 9 screens.*

Dataset	Tool-using agent	Pipeline (C)	Δ
Scharenberg22	**0.582**	0.465	+0.116
Carnevale22	0.044	**0.056**	−0.012
IFN-γ	0.163	**0.190**	−0.027
Sanchez21 (up)	0.059	**0.087**	−0.029
Sanchez21 (dn)	0.049	**0.099**	−0.050
Steinhart	0.100	**0.159**	−0.059
IL-2	0.264	**0.347**	−0.083
K562-GWPS	0.209	**0.347**	−0.138
K562-Essential	0.413	**0.556**	−0.143
Average	0.209	**0.256**	**−0.047**

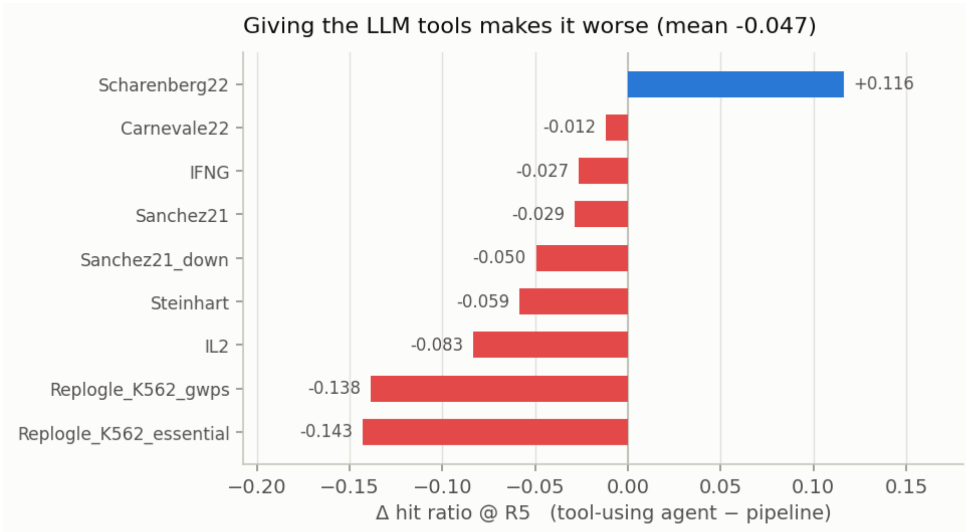

Figure 4: Per-screen delta of the tool-using agent against the pipeline. It wins on exactly one screen and loses worst precisely where the ML signal is strongest (K562-Essential, K562-GWPS) — given freedom, the LLM *overrides* a well-calibrated model.

The agent wins on exactly one screen (Scharenberg22, +0.116), where its runtime pathway-enrichment tool pays off, and loses most on the two genome-wide screens where the ML prior is strongest. The failure mode is instructive: given the freedom to act, the LLM overrides a model that is better calibrated than it is. **Constraining the LLM to a fixed fusion policy is not a limitation of the pipeline — it is the reason the pipeline wins.**

Transplanting the agent's one good idea

Since enrichment was the agent's only advantage, we transplanted it into the pipeline (Enrichr over the hits revealed so far, injected into the LLM prompt each round, gated to significant and coherent pathways), and separately tested an externalised *skill library* distilled from past experiments.

Table 6: Agent-style augmentations of the pipeline, as **paired** deltas over the same runs. Both are within noise of zero.

Augmentation	C	C + augmentation	Δ (paired)
Runtime pathway enrichment (gated)	0.260	0.263	+0.003
Externalised skill library	0.258	0.259	+0.001

Runtime enrichment captures the Scharenberg22 gain (+0.061 on that screen) without the collateral damage the free agent suffered, but nets only +0.003 overall: on genome-wide essentiality screens the enriched terms are highly significant yet *non-selective* (ribosome, proteasome, spliceosome), so "prioritise genes in these pathways" barely narrows a 9,000-gene pool. The skill library adds +0.001. Together with the memory ablation (+0.000), this is a consistent negative result about *externalisation*: memory, skills and retrieved pathways all add ≈ 0 once the model already has its parametric knowledge plus the hits it has observed.

Explainability: what is the LLM actually contributing?

The ablations say *how much* each component is worth; they cannot say *where* the value comes from. We therefore instrumented the arm to record, for every selected gene, a counterfactual attribution against the ML model's own top-k:

- **both** — the LLM named it *and* the ML would have selected it anyway;
- **LLM-only** — the LLM named it and the ML would *not* have selected it (this is the LLM's marginal contribution);
- **ML-only** — the LLM did not name it; it entered on ML score alone.

Pairing each pick with its revealed outcome gives the hit rate *per source* (9 screens $\times$ 5 rounds, $\approx$4,200 selected genes). K562-GWPS is reported separately: it is the one screen routed through the two-stage policy (the ML shortlists 384 candidates and the *LLM* makes the final selection), so the "ML-only" category cannot exist there and pooling it would corrupt the rates.

Table 7: Hit rate by attribution source, over the eight weighted-fusion screens. *Pooled* counts every selected gene once; *macro* averages the per-screen rates (screens with ≥ 10 picks in that category). Both aggregations agree on the headline: agreement between the two components is by far the strongest signal, and the LLM's unilateral picks are *not* better than the ML's.

Source	Genes selected	Hit rate (pooled)	Hit rate (macro)
ML + LLM agree	800	**21.1%**	**22.5%**
ML only	1,851	15.3%	13.4%
LLM only	1,509	10.2%	13.2%

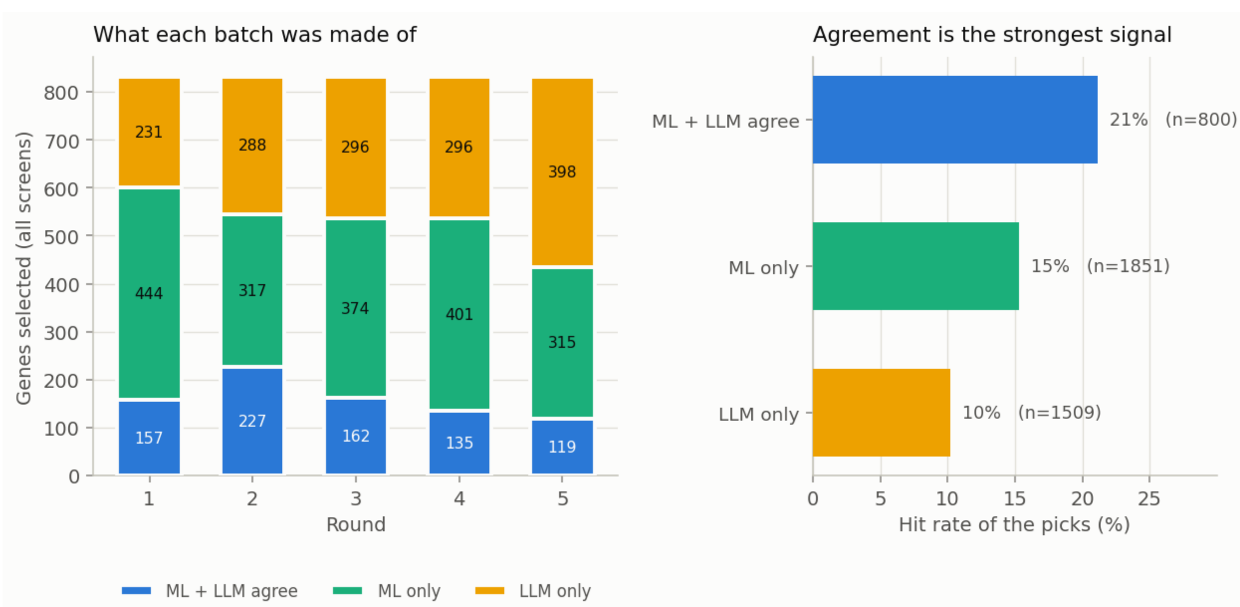

Figure 5: Left: what each round's batch is made of, pooled over the eight weighted-fusion screens. Right: the hit rate of each source. Genes that *both* components endorse hit at roughly twice the rate of genes the LLM introduces on its own.

The LLM is a verifier, not a proposer. Genes that both components endorse hit at 21.1% — about 1.4× the ML's own rate and 2× the rate of the genes the LLM introduces unilaterally. The LLM's *marginal* picks are no better than the ML's (worse when pooled, equal when macro-averaged, so the honest reading is "no better"). This single measurement explains the two results above that otherwise look unrelated:

- why **removing the LLM costs only** 0.008 — the genes it adds on its own are not better than what the ML would have picked instead; its value is concentrated in the *agreement* signal it provides over the ML's candidates;
- why **giving the LLM tools makes it worse** (−0.047, Table 5) — the free agent selects mostly on its own judgement, which is precisely the weakest source.

The design implication is the opposite of the prevailing "more agency" direction: the LLM should be used to *re-weight* a calibrated model's candidates, not to choose freely.

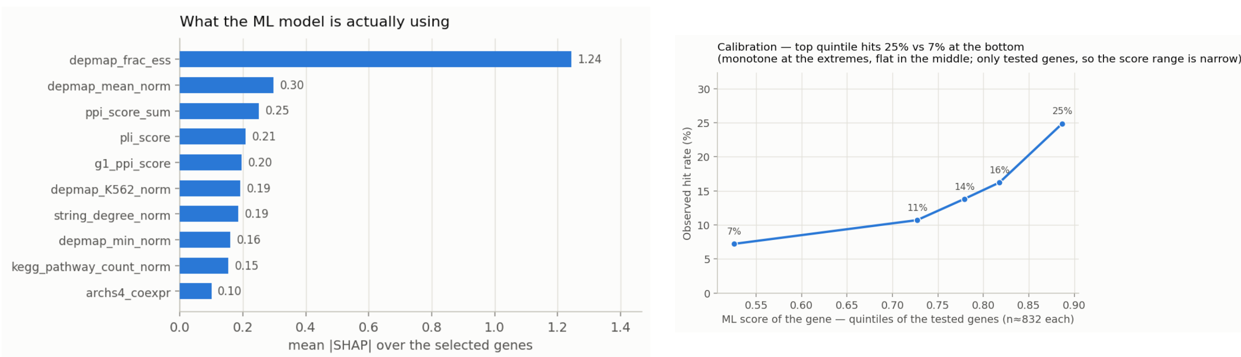

Figure 6: Left: features driving the ML score over the genes actually selected (LightGBM SHAP contributions; DepMap essentiality and PPI connectivity dominate). Right: calibration — the top-scoring quintile of tested genes hits about twice as often as the bottom quintile, though the curve is flat in the middle and the range is narrow because only *selected* genes are ever tested.